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 Studierichting: Informatica
 Oefeningenreeks 2D nr. 10.14

Opgave: Bereken de limiet $\lim_{x \rightarrow 2} \frac{x^3 - x^2 - 8x + 12}{3x^3 - 10x^2 + 4x + 8}$

Oplossing:

$$\begin{array}{r|rrrr} & 1 & -1 & -8 & 12 \\ 2 & & 2 & 2 & -12 \\ \hline & 1 & 1 & -6 & 0 \\ 2 & & 2 & 6 & \\ \hline & 1 & 3 & 0 & \end{array}$$

$$\Rightarrow x^3 - x^2 - 8x + 12 = (x - 2)^2(x + 3)$$

$$\begin{array}{r|rrrr} & 3 & -10 & 4 & 8 \\ 2 & & 6 & -8 & -8 \\ \hline & 3 & -4 & -4 & 0 \\ 2 & & 6 & 4 & \\ \hline & 3 & 2 & 0 & \end{array}$$

$$\Rightarrow 3x^3 - 10x^2 + 4x + 8 = (x - 2)^2(3x + 2)$$

$$\Rightarrow \lim_{x \rightarrow 2} \frac{(x - 2)^2(x + 3)}{(x - 2)^2(3x + 2)} = \lim_{x \rightarrow 2} \frac{x + 3}{3x + 2} = \frac{5}{8}$$

References